

A Hierarchical Intelligent Rehabilitation Robotic System Based on MI-EEG

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Abstract: In 2022, global stroke cases reached 101 million, posing a significant health threat. Traditional rehabilitation methods are primarily passive, while active recovery therapy leveraging brain plasticity proves more effective. Brain-computer interfaces (BCIs) establish a pathway between individuals and external devices. To address this, we have developed a robotic system that combines a BCI system with active rehabilitation. This system consists of two components: signals acquisition and classification, and robot control. For electroencephalography (EEG) signals, we use the Filter Bank Common Spatial Pattern (FBCSP) algorithm to extract spatial-frequency features and employ machine learning methods for classification. Additionally, we enhance classification effectiveness using EEGNet. The robotic system, divided into navigation, vision, and robotic arm modules, simulates patient-robot interaction through motor imagery (MI) tasks. Tests on five participants demonstrate EEGNet's superior classification accuracy (81.75% vs. 75.45%), with an overall task completion accuracy of 70.91%. The results show that the robotic system has a high accuracy in classifying MI-EEG, which is expected to create an intelligent rehabilitation system for future stroke patients.

Key Words: brain-computer interface, motor imagery, human-machine interaction, rehabilitation robot

1 Introduction

According to the World Stroke Organization (WSO) 2022, 101 million people worldwide were living with stroke aftermath [1]. There is a strong demand for stroke rehabilitation. Evidence-based medicine (EBM) confirms that active motor rehabilitation is the most effective way to reduce the disability rate.

Currently, BCI has attracted much attention in assisting the diagnosis and rehabilitation of limb movement disorders. BCI establishes a direct communication and control channel between the brain and external devices, bypassing peripheral nerves or muscles. It monitors brain activity and converts it into comprehensible commands for external device control or information transmission. A common method of signal collection is to place electrodes on the scalp or implant them in specific encephalic regions.

MI is a cognitive process of visualizing performing physical movements mentally. It activates the area of the brain responsible for performing actual movements, known as the motor cortex. As a result, MI is considered an effective training aid for improving motor skills and supporting rehabilitation.

At present, many researchers are dedicated to decoding the user's motor intention by processing and transforming EEG signals. Jin et al. [2] proposed a sparse Bayesian ELM-based method to classify EEG signals. Zhao et al. [3] employed spatial features from sampled electrodes and temporal features to create a 3D representation, which served as input for a multi-branch CNN. Although these methods have made great breakthroughs in classification, their effective-

ness is hindered by the monolithic nature of features. They may struggle to offer a high-level feature representation of complex EEG signal patterns.

Applying EEG signals to enable human-computer interaction is also a hot-point and frontier. Wang et al. [4] classified two states of imagined arm movement and relaxation, and controlled the MANUS robotic system. Cheng et al. [5] combined BCI with Soft Robotic Glove to enhance feedback during rehabilitation training and improve neuroplasticity effects. The above studies exemplify the potential application of BCIs in rehabilitation therapy, specifically in the development of rehabilitation robots.

In this study, we proposed a novel rehabilitation robot system based on MI-EEG detection. First, we took MI-EEG signals as input. Combining various dimensional features and utilizing neural network methods is anticipated to yield superior recognition results compared to single-dimensional feature. Furthermore, we developed an integrated BCI system for human-machine interaction and rehabilitation. The system not only includes an EEG signal acquisition module, but also incorporates rehabilitation robot navigation, vision and robotic arm modules, aiming to provide patients with more intuitive and personalized rehabilitation feedback.

The remainder of the work is divided into four sections. First, Section 2 describes the overall architecture and operating principles of the system. Then, Section 3 details the signal acquisition and classification methods. Section 4 introduces a rehabilitation robot system based on MI signals and presents experimental validation results. Finally, Section 5 summarizes this study and envisages the future work.

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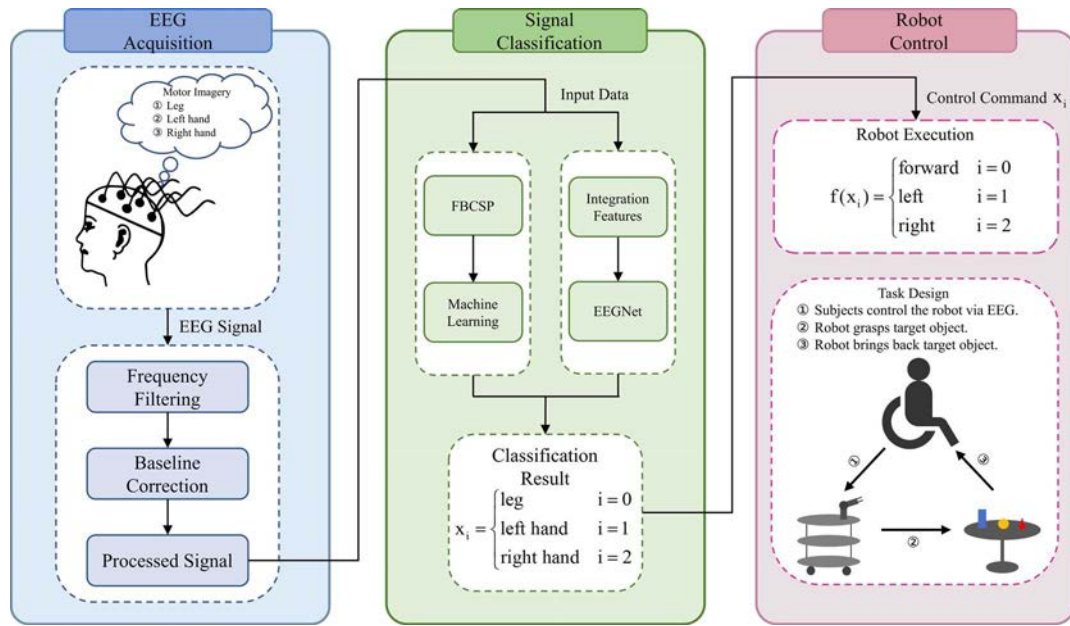


Fig. 1: The structure of the rehabilitation robot system

2 Overview

The specific structure of the rehabilitation robot system based on MI-EEG detection is shown in Fig. 1. It consists of three parts, which are the EEG acquisition module, signal classification module, and robot control module. In the EEG acquisition module, the subject's MI-EEG is extracted, pre-processed, and passed to the classification recognition module. The classification recognition module classifies the signals and passes the classification results to the robot control module to complete the designed task.

First, in the signal acquisition module, the subject receives instructions and imagines completing the corresponding actions. The EEG signals from specific channels of the subject are extracted with an EEG signal acquisition device. After frequency filtering and baseline correction, the signals are passed to the signal classification module.

Second, in the signal classification module, we classify the processed EEG data using machine learning and EEG-Net respectively, and transfer the classification results to the robot control module.

Finally, in the robot control module, we built a robot with autonomous navigation, visual recognition and grasping ability on the Robot Operating System(ROS) platform. It can do specific tasks according to the classification results of the subjects' EEG signals.

3 EEG Signal Acquisition and Analysis

3.1 Data and Experimental Paradigm

The entire dataset used was collected from experiments. Five subjects aged 20-22 years old (mean 20.4, variance 0.24) were recruited. We employed an 8-channel EEG amplifier (OpenBCI Cyton Board) to collect signals from the standard channels (Cz, C3, C4, FC3 and FC4) at a sampling rate of 256 Hz. The reference electrode(Ref) was placed in the earlobe of the right ear, while the ground electrode(GND) was on the left. The electrodes were set according to the international standard 10/10 lead system, as shown in Fig. 2(a).

Our study focused on the recognition of MI actions. Consequently, the following specific imagery tasks were selected: 1) leg movement; 2) left hand waving; 3) right hand waving. To ensure data homogeneity, we completed the acquisition in a coherent time period.

During the conduct of the experiment, subjects were in a quiet, well-lit room, seated comfortably approximately 0.5 meters away from a screen. They were instructed to stay focused, minimize eye movements, and avoid substantial body swaying. Each subject completed the corresponding imagery task by observing the prompts. The experimental paradigm is depicted in Fig. 2(b). When the computer emitted a beep sound, it indicated the start and the subject entered a preparatory phase. After 2s, the screen randomly generated textual prompts for the tasks along with images. Simultaneously, the subject commenced his imagery of the specified movements. The prompt continued to display for 6s and then disappeared, while the subject transitioned into a 2s rest phase.

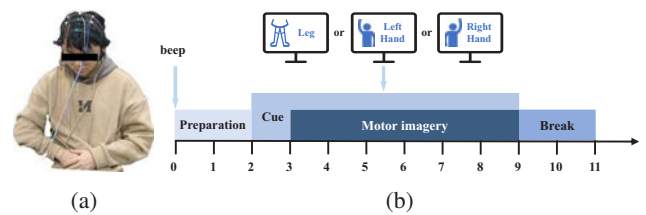


Fig. 2: EEG experiment design. (a) The subject and device; (b) The experimental paradigm.

The raw EEG signals were amplified and digitized, and then sent to the PC for preprocessing via the Bluetooth module. To facilitate rapid control response, a 3s time window with a 1s overlap was chosen. The time series data were segmented accordingly. Subsequently, we used a 5-40 Hz band-pass filter to eliminate the 50 Hz mains frequency interference. Meanwhile, baseline correction was implemented to retain essential information while refining the data.

3.2 Theoretical Basis for MI

Sensory Motor Rhythm (SMR) is a brainwave frequency property that occurs primarily during activity in sensory and motor areas [6]. Source imaging studies of SMR reveal that MI of different body parts is associated with a decrease in SMR in corresponding regions of the primary sensorimotor cortex[7].

During unilateral limb movement imagination, the contralateral sensorimotor cortex experiences a significant decrease in μ (8-13 Hz) and β (14-30 Hz) rhythm, termed event-related desynchronization (ERD). Simultaneously, the ipsilateral sensorimotor cortex shows an increase, known as event-related synchronization (ERS) [8].

3.3 FBCSP

EEG signals from different channels reflect the correlated information between different brain regions. ERD and ERS demonstrate the distinct neural processing dynamics in the left and right cerebral hemispheres when engaged in identical tasks. Therefore, information can be extracted from the SMR-modulated spatial patterns for classification purposes.

The FBCSP algorithm [9] utilizes mutual information and frequency decomposition techniques. It integrates the spatial pattern capturing capability of the Common Spatial Patterns (CSP) method [10]. This allows for the autonomous selection of discriminative temporal-spatial EEG characteristics.

It proceeds in four steps: frequency domain filtering, spatial filtering employing the CSP algorithm, feature selection, and classification.

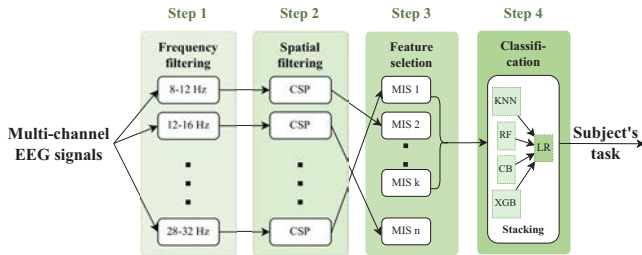


Fig. 3: Sequence of the FBCSP algorithm

Initially, we use zero-phase Chebyshev Type II filters to decompose the signals into 6 frequency bands within the 8-32Hz range. These bands cover μ and β rhythm, aiding in capturing relevant MI activities.

In the second step, spatial filtering is conducted in each frequency band. CSP, a mature spatial filtering technique for enhancing the distinctions between different categories, is well-suited for detecting ERD/ERS. The essence of the algorithm lies in finding the optimal orthogonal projection matrix $\mathbf{W}$ by diagonalizing the covariance matrices $\Sigma^{(+)}$ and $\Sigma^{(-)}$. This ensures maximum energy disparity in the spatial projection of EEG signals from various classes. The optimization targets the following objective function

$$J(\mathbf{W}) = \frac{\text{tr}(\mathbf{W}^T \Sigma^{(+)} \mathbf{W})}{\text{tr}(\mathbf{W}^T (\Sigma^{(+)} + \Sigma^{(-)}) \mathbf{W})} \quad (1)$$

where $\mathbf{W}$ is the CSP projection matrix. The rows of $\mathbf{W}$ are the stationary spatial filters and the columns of $\mathbf{W}^{-1}$ are the common spatial patterns.

Using the optimal common spatial filter $\mathbf{W}$, we convolve the measured signals $\mathbf{E}$ to obtain

$$\mathbf{Z} = \mathbf{W}\mathbf{E} \quad (2)$$

where $\mathbf{E}$ is an $N \times T$ matrix indicating the raw EEG data of a single trial; N is the number of channels; T is the number of measurement samples.

By computing the row-wise variance of the generated signal $\mathbf{Z}$, followed by logarithmic transformation and normalization, features are obtained. The m first and last rows of $\mathbf{Z}$ form the feature vector $\mathbf{x}_p$ given as

$$\mathbf{x}_p = \log \left(\frac{\text{Var}(\mathbf{Z}_p)}{\sum_{i=1}^{2m} \text{Var}(\mathbf{Z}_i)} \right). \quad (3)$$

The CSP method is inherently designed for binary classification problems. To address a three-categorization problem, we employ the One-vs-Rest (OVR) strategy. In OVR-CSP, each category is individually contrasted with the rest, to form multiple spatial filters. Discriminative features from diverse filter groups are fed into the classifier for training, utilizing the Mutual Information Based Feature Selection (MIFS) algorithm.

To overcome the limitations of a single model, we introduce the stacking integration strategy. It entails training a meta-model using outputs from multiple base models. The meta-model's role is to identify appropriate fusion mechanisms and mapping relationships for higher accuracy of final predictions.

Finally, we opt for 4 generalized models for training, including K-Nearest Neighbors(KNN), Random Forest(RF), eXtreme Gradient Boosting(XGB), and Categorical Boosting(CB). The predictions from them are fed into the Logistic Regression meta-model, which undergoes training via 5-fold cross-validation to yield the ultimate categorical predictions.

3.4 EEGNet

We also use EEGNet[11] as a classification network. EEGNet is a compact, multi-paradigm usable CNN with the following network structure shown in Fig. 4.

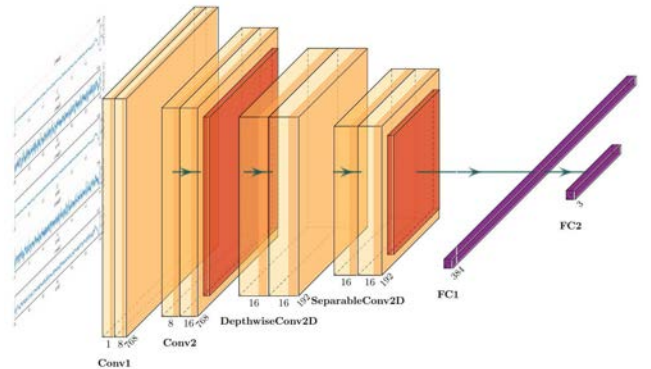


Fig. 4: EEGNet Network Architecture

This network is mainly divided into three parts, the first part is the INPUT and two ordinary convolutional layers, first of all, using F_1 temporal filters whose size is (1, 128),

and setting the length of the convolutional kernel to half of the sampling frequency can efficiently capture features above 2Hz.

The second part is the Depthwise convolutional layer which conducts convolution completely in the two-dimensional plane. Employing a channel-by-channel approach, it computes a convolution kernel with only one channel, leading to a significant reduction in parameters. Each feature map can learn D spatial filters. This layer refrains from performing weighted summation across different channels in the same spatial location. Due to its exclusive operation in the 2D plane, it cannot expand the number of feature maps. At the end, batch normalization is applied along the feature map dimensions, applying the ELU activation function and dropout with average pooling.

The ELU activation function is as

$$f(x) = \begin{cases} x \\ \alpha(e^x - 1) \end{cases} \quad (4)$$

where α is a hyperparameter, usually set to 1 shown in Fig. 5. It serves to scale the slope of the activation function for better handling negative inputs.

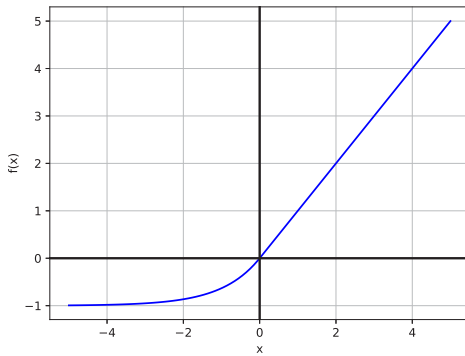


Fig. 5: ELU activation function ($\alpha = 1$)

ELU exhibits faster training time compared to the ReLU activation function. With the inclusion of negative values, ELU tends to center the mean of the activation output unit closer to 0, providing a regularization effect. And the negative value of ELU is calculated as an exponential function, which is more robust to input changes or noise and also more suitable for the case of EEG as raw data.

The third part comprises the Separable Convolutional Layer and the Classification Output Layer. The former consists of a combination of the Depthwise(DW) Convolutional Layer and the Pointwise(PW) Convolutional Layer. The 2D feature maps generated from the DW Convolutional Layer are input into the PW Convolutional Layer. The kernel size of the PW Convolutional Kernel is fixed as $1 \times 1 \times 3 \times F_2$, which will be used to regularize the received feature maps. This process combines the received feature maps in the depth direction with the same spatial weighting, which just makes up for the two shortcomings of Depthwise. This integration effectively reduces the number of network parameters.

3.5 Experimental Results

In the machine learning part, Table. 1 shows the signal classification accuracy for different subjects by using

single machine learning methods and the Stacking strategy. The table demonstrates that the introduction of the Stacking strategy improves the classification accuracy to some extent. However, the overall accuracy is relatively low. This may arise from the inter-individual variability. The FBCSP algorithm relies on manually designed filters, which are more demanding for selecting frequency bands. In addition, its adaptability to non-smooth signals is somewhat limited.

In the EEGNet part, we employed the Adam optimizer with a learning rate set to 0.0003. Specifically, $F1$, D , and $F2$ were chosen as 8, 2, and 16, and the number of epochs was set to 500. We utilized the cross-entropy loss function. The training results are presented in the Table. 1. It is evident that the classification accuracy is relatively satisfactory and varies across different subjects, ranging from a minimum of 77.09% to a maximum of 84.91%, with no instances of extreme deviation. The average classification accuracy is 81.75%, outperforming the FBCSP method. This improvement may be attributed to the neural network's ability to effectively learn the underlying features of the signal in both the time and frequency domains. Our system ultimately adopts the classifier trained by EEGNet.

Table 1: Accuracy of offline tests

Methods	CSP+ RF	FBCSP+ CB	FBCSP+ KNN	FBCSP+ Stacking	EEGNet
sub1	69.03%	56.60%	69.25%	68.24%	84.48%
sub2	71.99%	77.47%	86.03%	85.18%	84.91%
sub3	77.42%	65.87%	82.04%	83.81%	81.91%
sub4	65.40%	59.54%	67.28%	71.57%	80.34%
sub5	61.69%	56.33%	65.29%	68.47%	77.09%
Avg	69.10%	63.12%	73.98%	75.45%	81.75%

4 Rehabilitation Robot

4.1 Rehabilitation Robot Structure

As shown in Fig. 6, the robot consists of three modules, namely the navigation module, the vision module, and the robotic arm module. The navigation module controls the mobile localization of the service robot and performs path planning; the vision module recognizes the information of external objects and understands the environment; and the robotic arm module enables the robot to interact with external objects and complete specific tasks.

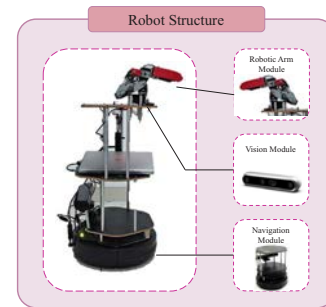


Fig. 6: Rehabilitation robot structure

1) Navigation Module

The navigation module enables the robot to adjust its position and direction according to the environment and plan a reasonable path based on its position and the target position.

It mainly includes three stages: map construction, localization, and autonomous navigation.

In the map construction stage, we use the gmapping software package to create 2D grid maps and use AMCL for localization. ACML algorithm solves the kidnapped robot problem in the MCL algorithm by utilizing KLD. By averaging the weights of all the particles ω_{avg} , and the long-term likelihood estimates ω_{fast} and short-term likelihood estimates ω_{fast} for each particle, which can be expressed as

$$\omega_{avg} = \omega_{avg} + \omega_t^{(i)} / N \quad (5)$$

$$\omega_{slow} = \omega_{slow} + \alpha_{slow} (\omega_{avg} - \omega_{slow}) \quad (6)$$

$$\omega_{fast} = \omega_{fast} + \alpha_{fast} (\omega_{avg} - \omega_{fast}) \quad (7)$$

where α_{slow} and α_{fast} are the average attenuation rates of the long-term and short-term exponential filters; N is the total number of particles.

In the AMCL algorithm, the probability of $\max\{0, 0, 1.0 - \omega_{fast}/\omega_{slow}\}$ is used to determine whether to add random particles [12].

In the navigation stage, the robot's visual sensor Kinect will be activated to collect information about the environment. The model of the robot will be displayed on the interface and the current coordinate and orientation of the robot will be calculated based on the sensor information.

In the path planning stage, the robot searches for the shortest path and adjusts the path dynamically according to the current environment by using the A* algorithm. The A* algorithm is a commonly used algorithm for path planning and graph search. Specifically, A* selects the path that minimizes the following equation

$$f(n) = g(n) + h(n) \quad (8)$$

where $f(n)$ is the total cost of the node; $g(n)$ is the actual cost from the start node to node n ; and $h(n)$ is the estimated cost from node n to the target node.

2) Vision Module

The vision module is divided into two aspects: image acquisition and target detection. For image acquisition, we use a Realsense D435i camera to acquire RGBD images, which can simultaneously acquire the color information and depth information of the object. For target detection, we use the YOLO network, which is a fast object detection algorithm that predicts the categories of all objects in an image and draws their bounding boxes. By combining color detection methods with the YOLO network, we can locate and identify multiple targets more accurately.

3) Robotic Arm Module

The robotic arm module consists of three AX-12 and two AX-18A servo motors controlled by the ArbotiX-M control board/USB2Dynamixel for target item grasping. In the item grasping task, the navigation module is first controlled to move toward the item position and align with the item. Then, the center coordinates of the item are acquired by the camera, and the required angle of each joint of the robotic arm is calculated using inverse kinematics.

4.2 Task Design

In the experiment, we simulated the scenario of a stroke patient recovering from a stroke and designed the task of the patient controlling the robot to complete the item grasping. As shown in Fig. 7, the patient can control the robot to move forward, turn left, and turn right by imagining leg, left hand, and right hand movements, respectively.

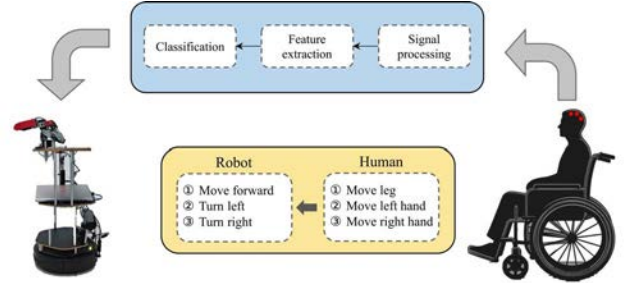


Fig. 7: Task design and human-computer interaction

The specific task flow is as follows. First, the robot is located at the starting position, and the human controls the robot to move towards the target item. When the vision module recognizes the target item, the robot will take over the control of the person and automatically complete the item-grasping task through the navigation module and vision module. When the robot completes the grasping, then the human controls the robot to return to the start. When the robot detects that it has returned to the starting position, it will automatically drop the target item.

This task vividly embodies the concept of hierarchical intelligence. Human intelligence is prominently featured when directing the robot's movements. Meanwhile, the autonomous grasping of target objects by the robot highlights the advantages of machine intelligence.

4.3 Experimental Results

In the experiment, we used OpenBCI to capture the EEG of subjects' MI. The EEG signals were transmitted to the computer via Bluetooth and input into the trained EEGNet network model to obtain the classification results of the EEG signals. The classification results were transmitted to the robot via network communication, and the robot completed the corresponding tasks according to the classification results. Fig. 8 shows the actual testing process.

For five subjects, the classification accuracy of the model is shown in Table. 2. With an impressive average accuracy of 70.91%, subjects can basically realizes the control of the robot to complete the set tasks by MI, highlighting the classification model's generalization capability. The experimental results show that the developed MI-BCI system is applicable for controlling robot movements and executing simple tasks.

Table 2: Accuracy of actual tasks

Subject	sub1	sub2	sub3	sub4	sub5
Acc	74.02%	74.01%	72.50%	69.67%	64.37%
Avg	70.91%				



Fig. 8: Actual mission process. (a) the robot is at the start position; (b) the subject controls the robot's movement through MI; (c) the robot detects the target object; (d) the robot performs object grasping; (e) the robot completes the object grasping; (f) the robot returns to the start position and passes the object.

5 Conclusion

The stroke stands as the most fatal disease affecting the central nervous system today, resulting in severe neurological damage. Some patients suffer from severe aftermath, which limits their self-care ability and imposes a heavy burden on both families and society. Currently, researchers are committed to helping patients in recovery by accurately decoding EEG signals and utilizing BCIs. Historically, research on stroke rehabilitation robotics primarily focused on training interventions, with the sole aim of facilitating the recovery of patients' motor functions.

In this work, we proposed a hierarchical intelligent rehabilitation robot system based on MI drawing upon the advantages of previous BCI systems. We integrated features and employed EEGNet to precisely identify motor intentions, converting them into control commands. In our system, controlling the robot's movement not only allowed for rehabilitation training but also assisted patients in performing various daily tasks. This integrated design facilitated the reactivation of physical functions, elevating the overall quality of life for patients.

The advantages of this work can be listed as follows:

1) **Accuracy of Recognition:** The deep neural network architecture applies time-domain signals to capture potential features and dependencies, enhancing the ability to accurately recognize intentions of movement.

2) **Diversity of Functions:** Modularized general-purpose robots possess capabilities in navigation, visual perception, and gripping. This not only enhances the patient's mobility but also increases their independence in daily activities.

3) **Convenience of Interaction:** The hierarchical intelligence system, with EEG data as the core control signal, seamlessly integrates human and machine intelligence. It enhances the man-machine integration performance and effectively relieves patients' fatigue during usage.

For future work, applying deeper neural network architecture for extracting fine temporal information is a feasible progression. In addition, the evolution of rehabilitation robots will integrate multi-degree-of-freedom and spatial intelligence, bridging the gap between executing complex tasks and enhancing rehabilitation outcomes.

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